Victor Jean Viaunel

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RESEARCH INTERESTS

Scan-to-BIM with mobile LiDAR · as-built BIM for existing buildings and energy retrofit · IFC and openBIM workflows · language models for BIM generation · computer vision for construction data

EDUCATION

2024 – 2027 **M.S. Civil Engineering** · National Taiwan University · in progress, expected Jan 2027
H-Team lab (BIM / IFC / Digital Construction), advised by Prof. Shang-Hsien (Patrick) Hsieh. Research area: scan-to-BIM with mobile LiDAR.

2019 – 2023 **B.S. Civil Engineering** · National Yang Ming Chiao Tung University

PUBLICATIONS

2026 · Proceedings **Victor, J. V.**, Kumar, P., & Hsieh, S.-H. (2026). From 2D drawings to BIM models: An LLM-assisted approach. In *Proceedings of the 43rd International Symposium on Automation and Robotics in Construction (ISARC)* (pp. 2410–2414). Singapore: IAARC. doi:10.22260/ISARC2026/0307

2026 · Conference Vignesh, V. P., **Victor, J. V.**, Senthilkumar, V., & Hsieh, S.-H. (2026). From scan to field: AR visualization of TLS-based as-laid utility geometry. *ASCE International Conference on Computing in Civil Engineering (i3CE 2026)*, Incheon. Proceedings forthcoming.

2026 · Conference **Victor, J. V.**, Chang, Y.-T., & Hsieh, S.-H. (2026). A review on scan-to-BIM for building energy retrofit. *International Conference on Computing in Civil and Building Engineering (ICCCBE 2026)*.

2025 Chang, J.-C., Yang, C.-T., **Victor, V. J.**, et al. (2025). Revisit public housing design in Taiwan: Automated design and evaluation workflow integrating Shape Machine and local data. *The 29th Symposium on Construction Engineering and Management / International Conference*.

PRESENTATIONS

Jun 2026 “From 2D Drawings to BIM Models: An LLM-Assisted Approach.” ISARC 2026, Singapore. Presenter.

25 Mar 2026 “A Review on Scan-to-BIM for Building Energy Retrofit.” ICCCBE 2026, Session 8B: Infrastructure Systems. Presenter.

2025 “Revisit Public Housing Design in Taiwan: Automated Design and Evaluation Workflow Integrating Shape Machine and Local Data.” The 29th Symposium on Construction Engineering and Management / International Conference. Presenter.

RESEARCH AND TEACHING

Oct 2024 – Present **Assistant Researcher** · H-Team lab, National Taiwan University
Research on scan-to-BIM and LLM-assisted BIM modelling, from point-cloud processing to IFC export.

Current

Teaching Assistant · BIM course, Prof. Shang-Hsien Hsieh, NTU

Support students with practical BIM topics and help them develop their own implementation ideas.

Jun – Aug 2022

Research Intern, Computer-Aided Engineering · National Taiwan University

Computer vision project in OpenCV and PyTorch, plus sequence models with LSTMs and GRUs.

PROFESSIONAL EXPERIENCE

Jun 2024 – Present

Lead Full-Stack Developer · Legend Castle, Taiwan

Main developer of MaiSchool, a live online tutoring platform for families, from design to production (Next.js, Convex, Clerk, Zod).

Feb – Jun 2023

Structural Engineering Intern · Envision

Assistant structural engineer; structural modelling and analysis in ETABS and SAP2000.

COMPETITIONS

2026

International AI Challenge on Marine Debris Image Recognition · Taiwan

15th of 69 teams in the preliminary round (two-person team), macro mAP@0.5 of 0.7917 across 34 debris classes.

2026

AI CUP: Solar Panel Defect Detection · Ministry of Education, Taiwan

Public score improved from 0.301 to 0.8337; about 29th of 121 teams.

SERVICE AND LEADERSHIP

Current · 2nd term

Academic Committee Lead · Haitian student community in Taiwan

Guide bachelor's to graduate applicants through Taiwanese university admissions: study plans, autobiographies, documentation, and application workshops that I coordinate.

SKILLS

BIM and engineering	Revit, IFC, AutoCAD, ETABS, SAP2000, SAFE
Reality capture	Mobile LiDAR scanning, point-cloud processing, photogrammetry
Computing	Python, PyTorch, OpenCV, YOLO, LLM agents, Next.js, Convex, MongoDB, Docker
Languages	English (fluent), Haitian Creole (native), Chinese (intermediate listening, basic speaking)